Information Technology University, Lahore

Assignment 1
BSSE23-A/B

Multivariable Calculus | (MT101T) | Fall ‘25

Name: ________________________ Reg. No.: ______________________

Total Points = 35

Q1. [CLO-1] [4pts]

Use vectors to find the fourth vertex of a parallelogram, three of whose vertices are $(0,0), (1,3)$, and $(2,4)$.

Q2. [CLO-1] [6pts]

In each part, find two unit-vectors in 2-space that satisfy the stated condition.

a. Parallel to the line $y = 3x + 2$
b. Parallel to the line $x + y = 4$
c. Perpendicular to the line $y = -5x + 1$

Q3. [CLO-1] [10pts]

a. Use vectors to show that $A(2,-1,1)$, $B(3,2,-1)$, and $C(7,0,-2)$ are vertices of a right triangle. At which vertex is the right angle?
b. Use vectors to find the interior angles of the triangle with vertices $(-1,0), (2,-1)$, and $(1,4)$. Express your answers to the nearest degree.

Q4. [CLO-1] [10pts]

Sketch the vectors $\boldsymbol{r}_0 = \langle -1,2 \rangle$ and $\boldsymbol{v} = \langle 1,1 \rangle$, and then sketch the six vectors $\boldsymbol{r}_0 \pm \boldsymbol{v}$, $\boldsymbol{r}_0 \pm 2\boldsymbol{v}$, $\boldsymbol{r}_0 \pm 3\boldsymbol{v}$. Draw the line $L: x = -1 + t, y = 2 + t$, and describe the relationship between L and the vectors you sketched. What is the vector equation of L?

Q5. [CLO-1] [5pts]

Find an equation of the plane through the point $(-1,4,2)$ that contains the line of intersection of the planes $4x - y + z - 2 = 0$ and $2x + y - 2z - 3 = 0$.